

Simulating Heat Transfer in a Closed Cavity

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Abstract

The objective of this project is to simulate heat flow in a closed cavity using OpenFOAM. A closed 2D cavity of dimensions 5 m by 5 m is chosen as the domain. As shown in Figure 1 below, a small portion BC of the bottom wall AD receives a constant heat flux q , while walls AB and CD are insulated. The remaining walls are kept at ambient temperature of 30°C. The length of BC is kept at 0.5 m, 1 m and 1.5 m, and transient natural convection is simulated with air as the working fluid inside, taking its properties at 30°C. The cavity is created using the `blockMesh` utility of OpenFOAM, and the convection is simulated using `buoyantPimpleFoam`. Results are obtained and post-processed in Paraview.

1 Introduction

Heat transfer in closed cavities is a key problem in thermal engineering, commonly driven by natural convection. This simulation studies the heat transfer behavior within an air-filled closed cavity using the `buoyantPimpleFoam` solver in OpenFOAM. The setup includes a heated wall, insulated surfaces, and ambient boundary conditions, creating buoyancy-driven flow. The goal is to analyze the temperature distribution, velocity fields, pressure variation, and heat transfer coefficients under these conditions.

2 Problem Statement

To simulate the flow of heat in a closed cavity by transient natural convection. In this case, a 2D cavity of dimensions 5 m by 5 m, as shown in Figure 1, is chosen as the working domain. The length of wall BC, which receives a constant heat flux, is varied through 0.5 m, 1 m, and 1.5 m, while walls AB and CD are insulated. The remaining walls are kept at ambient temperature. Air is chosen as the working fluid within the cavity, taking its properties at 30°C. Variations in temperature, pressure, velocity and heat transfer coefficients.

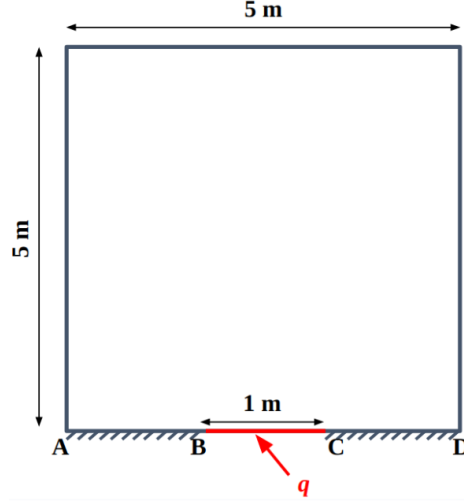


Figure 1: 2D Closed Cavity

3 Governing Equations and Models

The governing equations solved for this case involve compressible, buoyancy-driven flow with heat transfer. This reflects the natural convection that occurs within a closed cavity, where temperature gradients induce density variations in the fluid (air), resulting in motion driven by gravity.

3.1 Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0 \quad (1)$$

3.2 Momentum Equation

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \otimes \mathbf{U}) = -\nabla p_{rgh} + \rho \mathbf{g} + \nabla \cdot \boldsymbol{\tau} \quad (2)$$

where $p_{rgh} = p - \rho gh$ is the hydrostatic pressure and $\boldsymbol{\tau}$ is the viscous stress tensor.

3.3 Energy Equation

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{U} h) = \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \frac{Dp}{Dt} \quad (3)$$

where h is the specific enthalpy and α_{eff} is the effective thermal diffusivity.

3.4 Equation of State

$$\rho = \frac{p}{RT} \quad (4)$$

assuming ideal gas behavior.

Turbulence effects are modeled using the standard $k - \epsilon$ RANS model, which solves additional transport equations for turbulent kinetic energy (k) and its dissipation rate (ϵ). This approach captures the effect of turbulence on momentum and thermal transport without resolving the full turbulent spectrum.

4 Simulation Procedure

4.1 Geometry and Mesh

The geometry was created using the `blockMesh` utility of OpenFOAM. The thickness of the cavity was taken to be 0.01 m, and its $z = 0$ and $z = 0.01$ faces were set as type empty to make it a 2D case. The geometry was divided into three sections and meshed, so that the middle section contains the heated wall BC. The grading of the mesh is done using `simpleGrading`, and there was no refinement of the mesh carried out. The resulting mesh, as seen in Figure 2, has 7500 hexahedral cells.

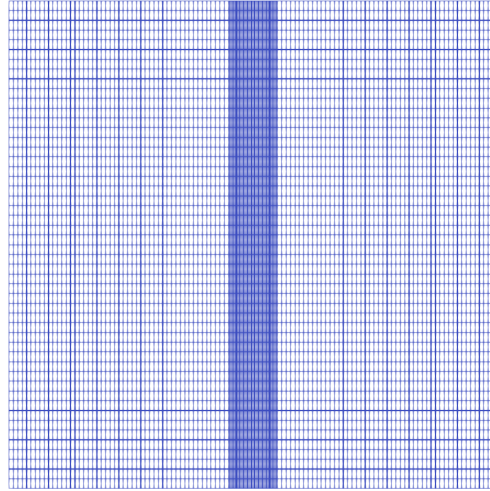


Figure 2: Generated Mesh

4.2 Initial and Boundary Conditions

All walls are treated as no-slip boundaries. Wall BC is assigned the `externalWallHeatFluxTemperature` condition, receiving a uniform heat flux of 1000 W/m^2 . Walls AB and CD are set to `zeroGradient`, effectively modeling insulated boundaries. The initial temperature inside the cavity is set to 30°C , which is also maintained as a fixed value on the remaining walls.

4.3 Solver

The simulation was performed using OpenFOAM's `buoyantPimpleFoam` solver, which is designed for transient, buoyancy-driven, compressible flows with heat transfer. It uses a combination of SIMPLE and PISO algorithms (referred to as the PIMPLE algorithm) to

handle pressure–velocity coupling efficiently in both steady and unsteady cases. This makes it particularly suitable for natural convection problems that involve temperature-dependent density variations.

5 Results and Discussions

The simulation was run for 30 s with a timestep of 0.02 s, and the following results were obtained.

5.1 Temperature Gradient

The temperature gradients for each of the three lengths of wall BC are as described in Figure 3.

The contours indicate that increasing the length of wall BC leads to greater heat transfer into the cavity, enhancing the strength of the thermal plume. The resulting mushroom-shaped plume is characteristic of buoyancy-driven flow and forms due to rising hot air entraining cooler surrounding fluid. This shape reflects the formation of vortex rings and upward momentum followed by lateral spreading as the plume decelerates. The symmetry of the plume suggests uniform heating and stable boundary conditions.

5.2 Velocity Contours

The velocity contours at $t = 30$ s for each length of BC is shown in Figure 4.

The x-velocity contour shows a symmetric horizontal flow pattern, with positive values on the right and negative values on the left, centered around the vertical axis of the plume. This indicates lateral inflow of cooler air at the bottom and outward spreading of heated air near the top. The y-velocity contours air rising in the middle and falling back down near the walls.

The velocity magnitude plot reveals that the strongest motion occurs along the central vertical axis, where the heated plume rises upward with high velocity. In contrast, the regions around the column (side walls and corners) show very low velocity magnitudes, indicating stagnant or weakly recirculating zones. This absence of significant movement around the column suggests that the buoyancy-driven flow is highly localized to the region above the heated wall segment.

5.3 Pressure Plot

The air pressure inside the cavity is plotted over time and is as shown below for each of the three cases.

The plots highlight the ranges of pressure in the cavity over time. The lighter shade denotes minimum and maximum pressures, the darker shades first and third quartiles, and

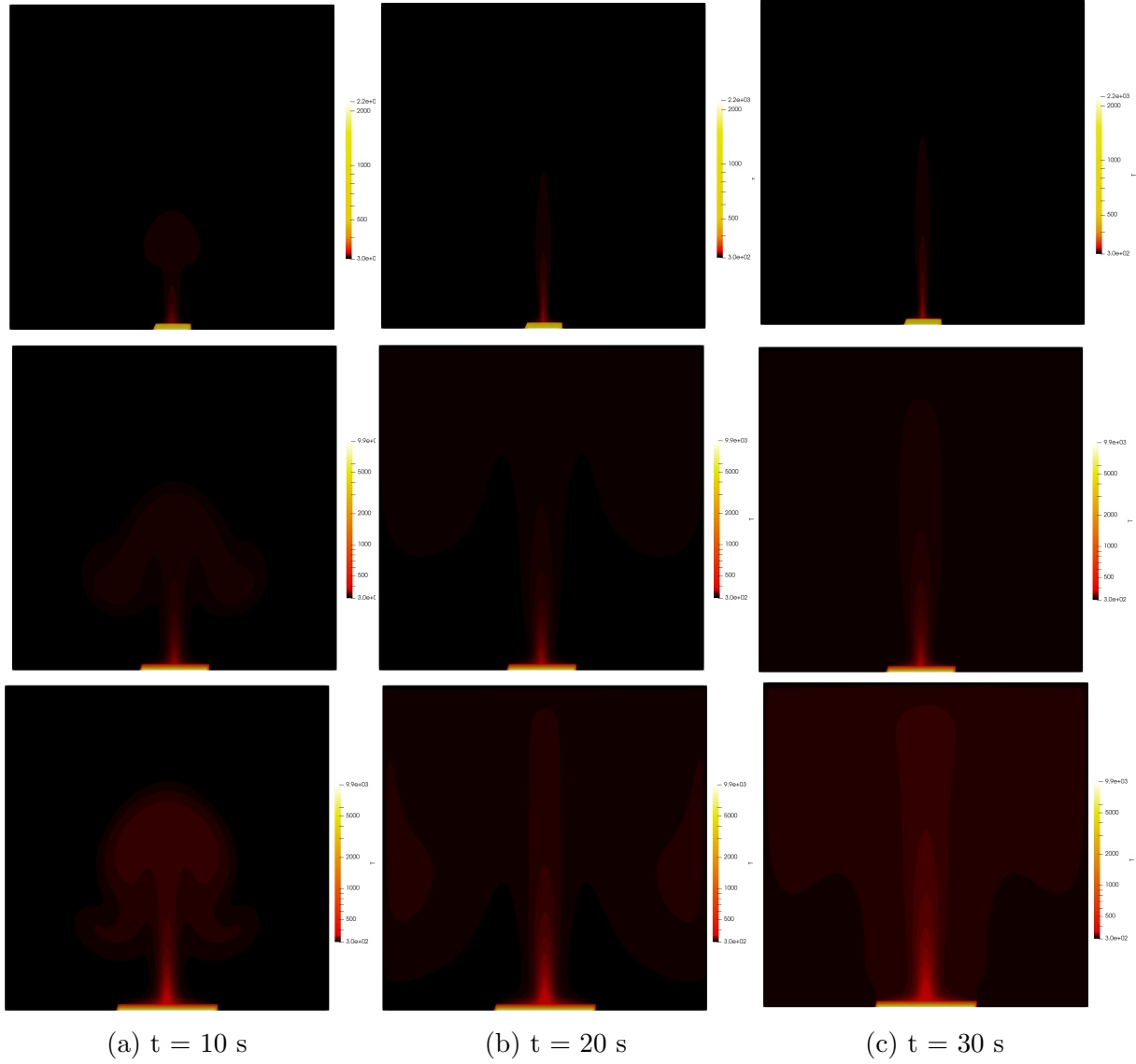


Figure 3: Temperature Gradients for BC = 0.5 m, 1 m and 1.5 m row-wise

the black line denotes average pressure. As we see, not only does the average pressure rises almost linearly, the extent to which it rises increases with increase in length of BC, due to more and more heating from the hot wall. Also, the variation of pressure decreases with increase in the length of BC. This suggests that a longer heated surface promotes stronger and more uniform upward flow, reducing localized pressure gradients. The plume becomes more stable and well-developed, leading to smoother pressure distribution and less fluctuation across the domain.

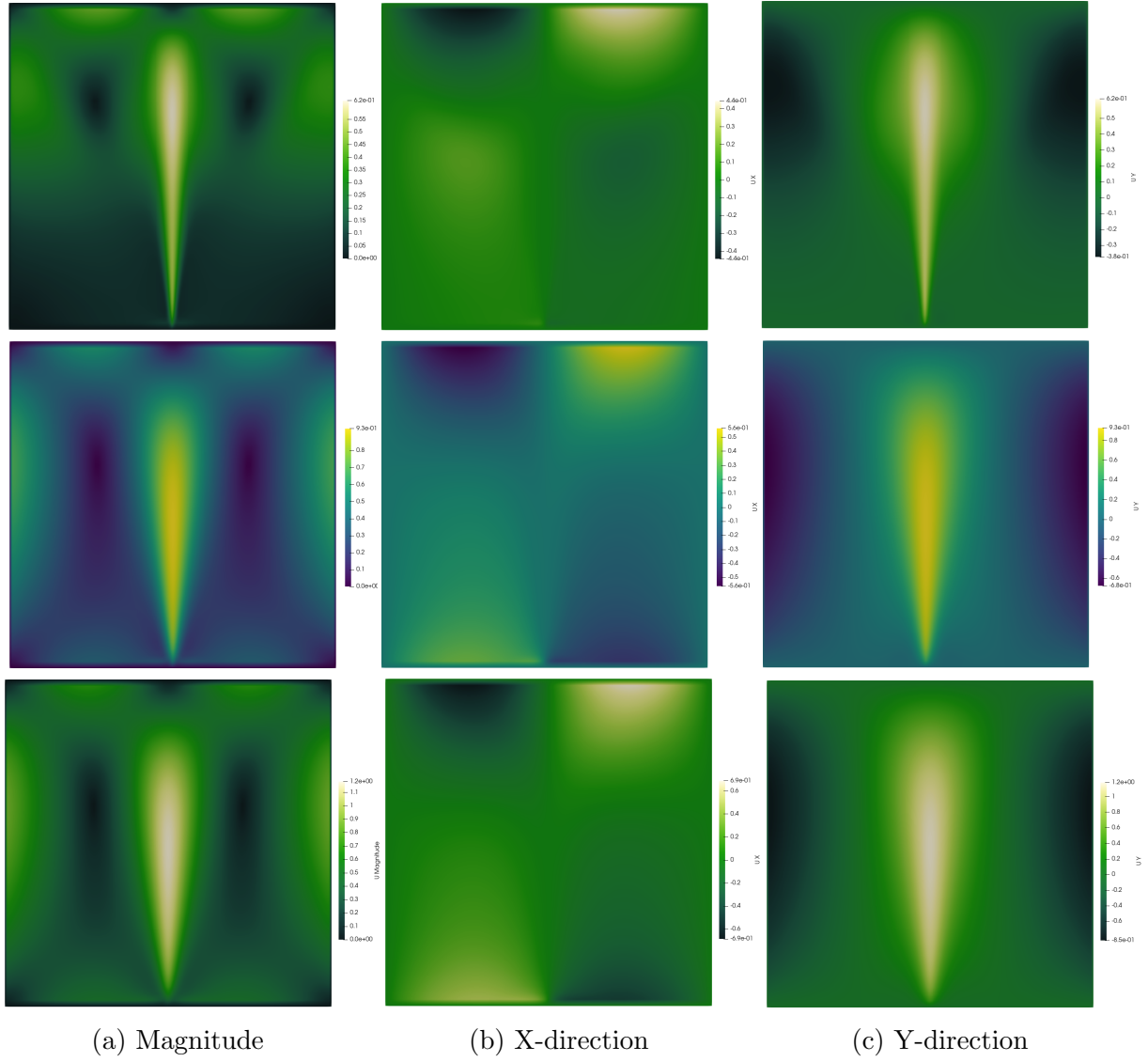


Figure 4: Velocity Contours for BC = 0.5 m, 1 m and 1.5 m row-wise

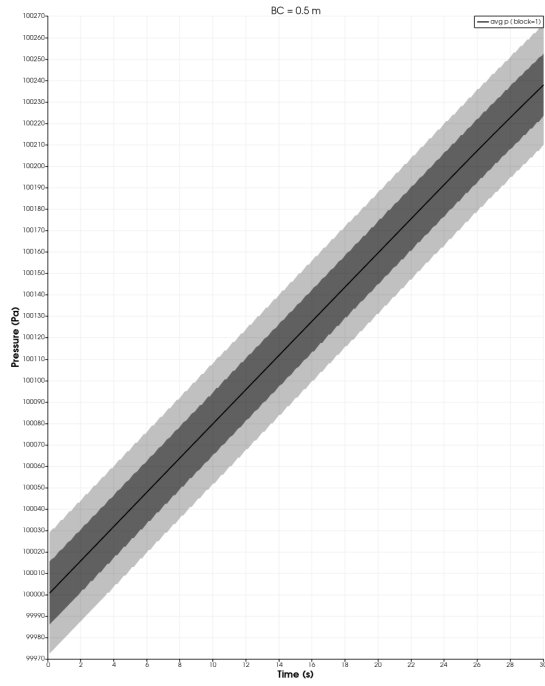
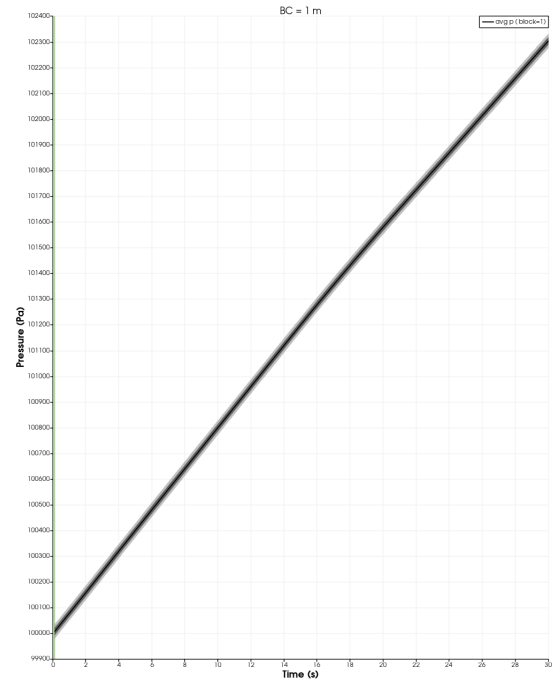
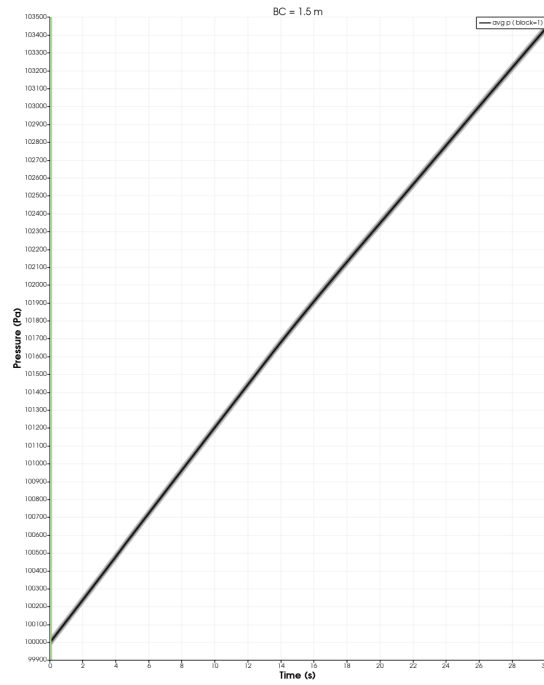
(a) $BC = 0.5 \text{ m}$ (b) $BC = 1 \text{ m}$ (c) $BC = 1.5 \text{ m}$

Figure 5: Pressure vs Time

6 Reference

- buoyantPimpleFoam Documentation
- blockMesh Manual